

BATESVILLE 8 WNW, AR, USA

WMO: 722450

Lat: 35.8201N

Lon: 91.7812W

Elev: 455

StdP: 14.46

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 23904

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.1	17.7	2.2	6.3	16.4	7.0	8.0	20.9	15.3	46.6	14.1	46.5	2.8	N/A	0.378

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.4	96.2	76.1	93.4	76.2	91.1	75.9	79.5	90.6	78.4	89.2	77.4	88.0	5.5	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.5	140.4	85.7	75.3	135.0	84.6	74.3	130.5	83.3	43.3	90.5	42.2	89.7	41.1	87.9	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.9	11.3	9.9	6.0	100.1	4.6	4.7	2.7	103.5	0.0	106.2	-2.6	108.8	-5.9	112.2		
(5)			4.4	81.4	4.8	1.1	1.0	82.2	-1.9	82.8	-4.5	83.4	-8.0	84.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.7	36.5	40.5	49.9	57.9	67.2	76.2	78.6	77.8	71.3	59.3	47.8	40.0		
	DBStd	17.01	10.42	11.00	10.14	8.37	7.12	4.72	4.87	5.10	6.23	8.53	9.35	9.27		
	HDD50	1395	435	306	129	24	2	0	0	0	0	16	148	335		
	HDD65	3875	883	688	472	239	65	0	0	0	16	219	519	774		
	CDD50	4571	16	40	128	262	534	787	886	862	640	306	84	26		
	CDD65	1575	0	1	5	27	133	337	421	397	206	43	4	1		
	CDH74	15897	0	5	64	221	1142	3521	4473	4010	1951	483	27	0		
	CDH80	6891	0	0	3	17	286	1551	2169	1945	793	125	2	0		
	WSAvg	3.8	4.8	5.2	5.2	4.8	3.5	3.2	2.6	2.7	2.7	3.1	4.0	4.3		
	PrecAvg	49.2	3.5	3.3	4.6	5.0	4.6	3.5	4.1	3.0	4.3	4.1	4.7	4.5		
(15) Precipitation	PrecMax	65.0	6.9	6.7	11.7	11.4	9.1	5.9	6.6	6.9	9.5	13.1	11.9	8.2		
	PrecMin	34.1	0.7	1.1	1.4	1.0	0.5	1.1	1.5	0.4	0.2	0.1	1.1	0.4		
	PrecStd	8.0	1.6	1.7	2.5	2.2	2.4	1.4	1.4	1.7	2.2	2.9	2.9	2.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.5	74.4	79.5	82.4	88.7	97.8	100.3	100.8	94.3	87.6	78.2	69.6		
		MCWB	59.7	62.4	64.0	68.6	72.1	74.8	73.9	78.0	74.9	70.6	63.2	62.0		
	2%	DB	64.4	69.1	75.7	79.4	85.7	93.3	95.9	96.5	91.3	83.8	72.3	66.2		
		MCWB	55.7	60.4	63.5	65.5	71.9	76.1	76.2	76.5	74.8	68.7	62.1	60.0		
	5%	DB	59.5	65.2	71.4	76.7	83.4	90.7	93.0	93.2	88.6	79.5	68.7	61.2		
		MCWB	51.6	56.1	59.6	64.8	71.5	75.6	77.2	75.8	73.8	65.5	60.0	54.9		
	10%	DB	54.8	60.4	67.5	73.6	80.7	88.2	90.5	90.0	85.0	75.5	64.6	56.3		
		MCWB	47.7	53.7	57.4	63.1	69.9	74.8	76.9	75.5	72.0	64.1	56.4	50.4		
	0.4%	WB	61.8	65.2	66.8	71.1	75.9	80.0	81.0	80.8	77.7	74.6	68.3	64.5		
		MCDB	66.5	72.8	77.4	78.8	84.8	90.8	91.2	92.5	89.8	84.5	74.8	68.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	57.6	62.2	64.2	68.8	74.4	78.3	79.8	79.3	76.4	72.0	64.7	61.2		
		MCDB	61.5	68.4	73.4	76.9	82.8	89.4	90.5	90.7	87.8	80.3	70.0	65.4		
	5%	WB	52.8	58.8	61.4	67.0	72.8	77.3	78.6	78.1	75.1	69.2	61.1	56.2		
		MCDB	57.3	63.3	68.3	74.5	80.8	88.2	89.3	88.7	85.8	76.0	67.4	60.3		
	10%	WB	47.9	53.7	58.9	65.2	71.2	76.2	77.6	76.9	73.8	66.5	57.4	51.5		
		MCDB	54.8	58.7	64.7	72.2	79.1	86.6	88.1	86.7	83.6	73.5	63.1	55.0		
	5% DB	MDBR	21.5	21.1	22.8	23.3	21.2	21.7	21.4	21.7	23.0	24.9	23.5	20.2		
		MCDBR	27.8	26.6	27.1	25.7	23.8	24.7	24.3	26.2	26.5	28.1	26.5	25.2		
		MCWBR	20.5	19.8	17.8	15.1	12.5	10.9	9.1	9.6	11.6	14.9	17.4	19.0		
		MCWBR	22.0	22.9	24.3	20.0	20.2	21.4	21.0	21.7	23.6	24.0	21.0	20.4		
(35) Mean Daily Temperature Range	5% WB	MCWBR	18.8	18.1	16.8	13.6	11.8	10.3	9.4	9.6	10.9	15.0	16.2	17.4		
(40) Clear-Sky Solar Irradiance	taub		0.314	0.322	0.357	0.389	0.430	0.453	0.485	0.474	0.408	0.355	0.325	0.308		
	taud		2.487	2.463	2.399	2.307	2.253	2.208	2.164	2.172	2.346	2.475	2.513	2.521		
	Ebn at Noon		281	292	289	284	272	265	255	256	269	276	275	277		
	Edh at Noon		26	30	35	41	44	46	47	46	37	30	26	24		
(44) All-Sky Solar Radiation	RadAvg		771	945	1299	1640	1802	2062	1973	1813	1545	1178	835	663		
	RadStd		67	122	97	117	146	140	102	116	118	146	90	75		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (24 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page